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Cyber-Physical Threat Intelligence and Automated Incident Response for Critical Subsea Infrastructure

The physical backbone of the internet is the submarine fiber optic cables that carry over 97% of the global internet's digital traffic. They also facilitate roughly $10 trillion in daily financial transactions. Regardless of this criticality, these cables are basically left unguarded on the seabeds. This exposes to us a massive blind spot that we have in modern-day cybersecurity: local network defenses operate in total separation from the physical domain, especially when it comes to the maritime domain. The framework I have proposed, which I called Aegis-Sea, will look to bridge this gap between the physical domain and the cyber domain by functioning as a real-time threat monitoring and automated incident response engine. To construct this project, specifically with its elements of having to come up with the best way to do threat scoring and routing, we need all the external research data we can get. I read through and synthesized research papers from five different subjects to help me with threat modeling, automatic identification systems and their vulnerabilities, anomaly detection, spatial risk, and SDN resilience.

As we have already stated, the fragile nature of the global internet infrastructure is seen in the forensics of past alerts. In order for us to understand this kind of physical vulnerability, I read through the report by Sunak et al., which is titled "Undersea Cables: Indispensable, Insecure and Vulnerable to Hybrid Warfare." What this paper does is highlight that these vital cables, which, fun fact, are often not even thicker than an average garden hose, are publicly marked on nautical charts and exposed to daily maritime traffic. In this research, Sunak advocates for companies and governments to adopt cable protection zones in order to mitigate the primary threat of anchor dragging. We have even seen recent geopolitical events that validated Sunak's threat model, such as in another report titled "Protecting Maritime Infrastructure from Hybrid Threats" and in "Russia's Hybrid Threat Tactics Against the Baltic Sea Region." These kinds of papers highlight to us the grey zone within the Baltic Sea by highlighting incidents such as the Newnew Polar Bear (October 2023) incident and the Yi Peng (November 2024) incident. These carrier ships dragged their anchors over 100 nautical miles over the cables, severing the Balticconnector pipelines and multiple other communication cables. Reading these papers helped provide the geopolitical context for the problems Aegis-Sea will be tackling. Based on Sunak's recommendations in policy, my framework will also try to enforce a strict 3 nautical mile cable protection zone and a 9 nautical mile alert area around all the 712 global subsea cables. To go further, the forensic data from Sari and Praks' papers provide us with the exact kind of parameters for the framework's speed anomaly, anchor drop scores, and anchor dragging measurements.

While my framework will be relying on live vessel telemetry data, the maritime tracking protocols can actually possess critical security flaws, as I have learned. This paper titled "A Security Evaluation of AIS Automated Identification System," by Balduzzi et al., discussed exactly that. The paper's overall goal was to perform landmark security audits of AIS protocols. They demonstrated to us that the AIS operates over VHF radio with zero encryption and zero

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authentication and even no replay protection. Utilizing inexpensive software to create radios, they were able to successfully create and execute ghost ships. This unreliability is actually shown in another paper by Winkel et al., whose paper "SeaSpoolFinder" developed an algorithm to filter physically impossible speed jumps that can be seen as GPS spoofing in contested regions. In order to combat these kinds of vulnerabilities, the paper written by Scoldi et al. proposes to us a kind of sensor in their research "Space-Based Global Maritime Surveillance" which cross-references satellite AIS with non-cooperative SAR in order to find dark or suspicious vessels. My project will utilize Balduzzi's research, which proves that AIS cannot be blindly trusted, and integrate a multi-factor approach. Aegis-Sea will try to implement a specific kind of penalty for transponder blackouts and use Winkel's work to filter out impossible coordinate jumps of ships during telemetry injection.

A very important thing for our project and framework to get right is the ability to be able to distinguish malicious anchor grabbing from normal operations. This, of course, requires some kind of kinematic modeling. Using the paper "GeoTrackNet," which introduced a deep learning framework that utilized VRNN on over 4.2 million AIS messages off the coast of France. Through their research, they found that simple single-threshold proximity alerts generated massive false positives in busy shipping lanes. In order to fix this, they used NFA statistical bounds. In their paper, Capobianco et al. further discussed kinematic tracking by using mean-reverting processes, which are values that rubber-band but end up being the mean value at the end, in order to model the normal equilibrium of vessel speeds. My framework will adopt the thesis of Nguyen, where he stated how proximity is usually insufficient for alerts. Aegis Sea will also utilize multiple factors in its model, which will include things like proximity combing, grading, and dwell time limits so we can eliminate any false alarms. Evidently, in Capobianco's paper, it mathematically justifies penalizing merchant vessels that drop from that 15-knot cruising equilibrium to less than 4 knots.

While all these papers provide us with foundational models for their own individual components of maritime and network security, their findings remain separated. My framework will be different, as it will be a synthesis of all these ideas and bridge that gap between the physical maritime tracking and the logical infrastructure resilience. We see how existing frameworks like GeoTrackNet are really entirely focused on detecting general shipping anomalies or preventing ship collisions. But they are not designed to be a static monitor for vessels. We also see how papers with projects such as Nautilus and OpenFlow map the physical-to-logical relationship pretty well, but they are not reactive. These systems only execute failover protocols after a physical cable has already been severed and an incident has already occurred. My system, on the other hand, will unify the cyber and physical framework in order to bridge real-time vessel behaviors to actual internet incident responses. Rather than us waiting for a cable to just snap and the internet outages to occur for everyone to then finally do something,

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my system will evaluate the telemetry data live and alert the necessary people before the incident occurs.

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